Data Analytics Capstone

Machine-Learning Forecasting of Solar Irradiance and Reliability-Constrained Off-Grid Solar Sizing Across Ghana's Climatic Zones

Synopsis

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Introduction

Background and Context

Ghana's electricity supply has traditionally relied on hydropower from the Volta basin and thermal generation fueled by imports, leaving the grid vulnerable to fluctuations in rainfall and fuel prices. The Government of Ghana's Renewable Energy Master Plan, adopted in 2019, set a target to increase renewable generation capacity from a 2015 baseline of 42.5 MW to approximately 1,364 MW by 2030, including about 500 MW of utility-scale solar and 200 MW of distributed solar photovoltaics (PV; Energy Commission of Ghana, 2019). Ghana's tropical climate provides abundant solar energy, with daily global horizontal irradiance (GHI) of approximately 4–6 kWh/m²/day and a distinct north–south gradient that favors the savannah regions (Asumadu-Sarkodie & Owusu, 2016).

Two groups would benefit substantially from accurate resource forecasting. First, grid operators and developers, including the Ghana Grid Company (GRIDCo) and the Electricity Company of Ghana (ECG), need short- and medium-term irradiance forecasts for generation planning, reserve capacity, and plant siting. Second, households, often overlooked in research, are increasingly adopting off-grid solar in response to unreliable grid power and falling PV and battery costs, yet they lack an evidence-based method for sizing installations. Installers commonly apply rules of thumb, such as a fixed number of days of autonomy, that ignore regional variation, for example, the elevated risk of consecutive low-sun days in coastal Accra during the rainy season compared with Navrongo in the Sudan savannah during the Harmattan.

Machine-learning (ML) methods now dominate the international solar-forecasting literature (Ahmed et al., 2020; Lauret et al., 2015; Voyant et al., 2017), and deep-learning models such as long short-term memory (LSTM) networks have achieved 18–34% error reductions over baselines in tropical settings (Qing & Niu, 2018). Attention-based architectures such as the Temporal Fusion Transformer, Informer, and PatchTST have also been benchmarked for irradiance forecasting recently (Shan et al., 2025); they are acknowledged here to situate the study within the current state of the field, although the present design retains established, more interpretable models suited to daily-resolution data. For Ghana specifically, Quansah et al. (2022) evaluated NASA Prediction of Worldwide Energy Resources (POWER) solar-radiation estimates against ground-station-based estimates derived from sunshine-duration observations at 22 Ghanaian synoptic stations, reporting correlations of approximately r = .60–.94 and supporting NASA POWER as a long-term solar data source for regional comparison. Because correlation alone does not establish unbiased agreement, site-level bias and uncertainty will be examined explicitly during exploratory analysis. Ghana-specific ML research on solar forecasting remains limited to plant-level studies such as Asiedu et al. (2024), and the review conducted for this proposal identified no published work linking forecasting-grade irradiance modeling to household system sizing.

Problem Statement

Solar resource uncertainty in Ghana is currently managed with simple averages and heuristics: Grid planners lack systematic evidence about which forecasting models are most effective across climatic zones and seasons, and off-grid households size panels and batteries with fixed rules that can waste capital through oversizing or cause blackouts through undersizing. This project addresses two gaps. First, the preliminary literature review identified no published study that systematically compares classical and ML irradiance forecasts across Ghana’s coastal, forest, and savannah zones using a common long-term dataset and formal forecast-comparison tests. Second, the review identified no publicly documented, reproducible tool that converts Ghana's historical irradiance record into reliability-constrained, zone-specific off-grid sizing guidance, i.e., the number of panels and the battery capacity needed to meet a specified consumption level at a given reliability level year-round. The review searched Google Scholar, Scopus, ScienceDirect, and IEEE Xplore using combinations of "Ghana," "solar irradiance forecasting," "machine learning," "NASA POWER," "off-grid PV sizing," and "loss-of-load probability" for publications through July 2026; the search will be repeated and documented before the final report so that the gap statements remain auditable.

Purpose of the Study

The study pursues two related aims: prediction and optimization. First, it predicts daily GHI by developing and statistically evaluating classical models (persistence, day-of-year climatology, and SARIMA-class dynamic harmonic regression) and ML models (random forest, XGBoost, and LSTM) at five locations spanning Ghana's three climatic zones, using 20 years (2005–2024) of NASA POWER daily data, and it explains the meteorological drivers of irradiance using Shapley additive explanations (SHAP). Second, it optimizes by transforming the same 20-year record into a reliability-constrained off-grid sizing model that identifies the smallest feasible PV array and battery configurations that satisfy predefined reliability constraints through historical energy-balance simulation. The contribution is a reproducible decision-support framework that combines ML forecasting, historical simulation, and reliability-constrained optimization to generate zone-specific PV–battery sizing recommendations for Ghana.

Scope and Objectives

These bounds translate into four measurable objectives, each paired with a research question below:

Objective 1 (O1). Train and evaluate six forecasting models per site and demonstrate statistically significant skill (Skill > 0) over the classical baselines on the untouched 2023–2024 test window at both the 1-day and 7-day horizons.

Objective 2 (O2). Rank the five meteorological drivers of daily GHI using correlation tests, permutation importance, and SHAP values, reporting Holm-corrected significance for each driver.

Objective 3 (O3). Quantify wet–dry-season and cross-zone differences in forecast accuracy using formal two-sample and multigroup tests at $\alpha = .05$.

Objective 4 (O4). Produce zone-specific PV–battery sizing tables that meet LOLP targets of 1% and 5% for the 100, 200, and 300 kWh/month consumption tiers.

The project is bounded to five study sites representing Ghana's climatic zones, i.e., Accra (coastal savannah), Takoradi (coastal), Kumasi (moist forest), Tamale (Guinea savannah), and Navrongo (Sudan savannah), at daily resolution over 2005–2024, with forecast horizons of 1 day and 7 days ahead. Sub-daily forecasting, plant-level power telemetry, and primary data collection are out of scope. The five sites were selected because together they span Ghana's three recognized climatic zones and the full north–south irradiance gradient; because NASA POWER provides spatially smoothed gridded data, additional sites within the same zone would contribute little independent climatic information. Horizons follow a direct forecasting strategy: a separate model for each horizon predicts GHI at day $t + 1$ or day $t + 7$, and all predictors are restricted to information available at the forecast origin, t. The forecasting comparison (Research Questions 1–3) is the primary contribution and the sizing application (Research Question 4) the secondary contribution; this prioritization protects the study if time becomes constrained. The study pursues four research questions:

Research Question 1 (RQ1). How accurately can machine-learning models (random forest, XGBoost, LSTM) forecast daily global horizontal irradiance at representative sites across Ghana's coastal, forest, and savannah climatic zones, and do they significantly outperform classical approaches (persistence, day-of-year climatology, and dynamic harmonic regression) at the 1-day and 7-day forecast horizons?

Research Question 2 (RQ2). Which meteorological variables (temperature, relative humidity, cloud amount, precipitation, wind speed) are most strongly associated with daily solar-irradiance variation in Ghana, and which contribute most to predictive performance?

Research Question 3 (RQ3). How does forecasting accuracy differ between the wet and dry seasons (including the Harmattan) and across the north–south climatic gradient?

Research Question 4 (RQ4). What is the minimum feasible PV-array and battery configuration that satisfies predefined reliability constraints (loss-of-load probability [LOLP] $\leq$ 1% and $\leq$ 5%) for representative household electricity demands across Ghana's climatic zones year-round?

A representative household electricity demand of 200 kWh/month benchmark will be used in RQ4; it represents a moderate-to-high household consumption scenario rather than the national median. Energy Commission of Ghana (2025) statistics indicate residential consumption of roughly 8,569 GWh across 5,281,816 residential customers in 2024, approximately 135 kWh per customer per month. Because customers are metered connections that compound houses often share, per-household consumption is likely lower still. Sizing results are therefore reported for the 100, 200, and 300 kWh/month tiers to span from low-consumption rural households to high-consumption urban households and to demonstrate sensitivity to the assumed demand level.

Sample Size Calculation

The dataset is a fixed archive rather than a survey to be fielded, so this section assesses whether the planned data volume is adequate for each research question, using confidence-interval (CI) precision and power-of-test methods as appropriate. Because daily observations are serially dependent, these independent-observation formulas serve as approximate benchmarks for precision and power; final uncertainty estimates and hypothesis tests use time-series-aware procedures (moving-block bootstrap; corrected Diebold–Mariano tests) that account for the reduction in effective sample size due to autocorrelation. Throughout, the significance level is $\alpha = .05$ ($z = 1.96$) and, where power is used, power is 90% ($z\beta = 1.28$). Table 1 summarizes the calculations.

These assumptions are anchored in prior empirical work rather than chosen freely. The 90% power criterion is deliberately stricter than the conventional 80%, and $\rho = .10$ and $d = 0.10$ correspond to Cohen's (1988) thresholds for small effects, so even weak meteorological drivers and modest accuracy gains remain detectable. The margin of error $E = 0.05$ kWh/m²/day is approximately 1% of Ghana's mean daily GHI of 4–6 kWh/m²/day, and $\sigma = 1.0$ kWh/m²/day approximates the day-to-day GHI variability observed at Ghanaian stations (Quansah et al., 2022). The paired effect size $d = 0.10$ is also far smaller than the error reductions of 18–34% reported for deep-learning forecasts in tropical settings (Qing & Niu, 2018) and the gains reported in machine-learning benchmarking studies (Lauret et al., 2015), so the design remains adequately powered even if the realized improvements are much more modest.

RQ1a (CI precision). To estimate mean forecast error with a margin of error $E = 0.05$ kWh/m²/day, taking $\sigma = 1.0$ kWh/m²/day as a deliberately conservative bound on error variability: $n = (\frac{z \times \sigma}{E})^2 = (\frac{1.96 \times 1.0}{0.05})^2 \approx 1{,}537$ daily observations per site.

RQ1b (power of test, paired comparison). Models are compared on the same test days, a paired design on daily error differences. To detect a small standardized effect, d = 0.10: $n = (\frac{z_{\alpha/2} + z_{\beta}}{d})^2 \approx 1{,}051$ paired test days.

RQ2 (power of test, correlation). To detect a weak correlation of ρ = .10 at α = .05 with 90% power via the Fisher z transformation, where C(ρ) = 0.1003: $n = (\frac{z_{\alpha/2} + z_{\beta}}{C(\rho)})^2 + 3 \approx 1{,}047$ observations. The archive is fixed, and all available observations will be used; this benchmark confirms adequacy rather than determining collection.

RQ3 (power of test, two-sample). To compare wet-season and dry-season forecast errors with effect size d = 0.20: $n = 2(\frac{z_{\alpha/2} + z_{\beta}}{d})^2 \approx 526$ days per season group.

RQ4 (CI precision, proportion). The LOLP is a proportion (the share of observed historical days on which the modeled system would fail to serve the load). To estimate an LOLP near p = .01 within E = .005 at 95% confidence: $n = \frac{z^2\, p(1-p)}{E^2} \approx 1{,}522$ historical days evaluated per zone–configuration. Because loss-of-load days occur in runs rather than independently, interval estimates for the LOLP use year-block and moving-block bootstrap methods.

Table 1

Minimum Sample Size per Research Question

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
RQ1a	Confidence interval (mean)	z = 1.96, σ = 1.0 kWh/m²/day, E = 0.05	1,537 days/site
RQ1b	Power of test (paired difference)	α = .05, power = .90, d = 0.10	1,051 test days
RQ2	Power of test (correlation, Fisher z)	α = .05, power = .90, ρ = .10	1,047 days

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
RQ3	Power of test (two-sample)	α = .05, power = .90, d = 0.20	526 days/season
RQ4	Confidence interval (proportion)	z = 1.96, p = .01, E = .005	1,522 days/zone

The final sample size is N = max(RQ1–RQ4) = 1,537 daily observations per site (about 4.2 years of daily data). The project retrieves 20 years (2005–2024) of daily data, approximately 7,305 observations per site and 36,525 in total across the five sites, exceeding the binding requirement by more than 4 times. For RQ1b, the held-out test period provides about 730 paired days per site, below the 1,051-day benchmark for a single site; the requirement is met by pooling test days across the five sites (about 3,650 paired observations), with per-site comparisons reported descriptively. The RQ3 per-season requirement is likewise met by pooling across sites (about 1,825 test days per season group, compared with the 526 required), and the full 20-year record supports the RQ4 historical backtest across all five sites simultaneously. Throughout, pooling across sites is used only to increase inferential precision; site-level estimates remain the primary substantive results. Because weather is spatially correlated across sites, pooled tests additionally resample time blocks jointly across all sites, so cross-site dependence is preserved in the resampling scheme rather than being assumed away.

Data Description

The primary data source is the NASA POWER archive (NASA Langley Research Center, 2024), which provides daily satellite-derived solar and meteorological parameters on a global grid, free of charge and without registration, via a documented Representational State Transfer (REST) application programming interface (API). Quansah et al. (2022) validated NASA POWER solar radiation data against ground-based estimates at 22 Ghanaian synoptic stations (r

= .60–.94), with the strongest agreement in the northern savannah, supporting their use in this study. Data are retrieved for the five study sites listed in Table 2, and Figure 1 shows a sample of the raw daily record as retrieved for Accra.

Data source URLs:

- Archive and documentation: https://power.larc.nasa.gov/
- Example API call (Accra, all six parameters, 2005–2024): https://power.larc.nasa.gov/api/temporal/daily/point?parameters=ALLSKY_SFC_SW_DWN,T2M,RH2M,CLOUD_AMT,PRECTOTCORR,WS2M&start=20050101&end=20241231&latitude=5.60&longitude=-0.19&community=RE&format=CSV

Figure 1

Sample of the Raw NASA POWER Daily Record for Accra (2005–2024)

```
POWER_Point_Daily_20050101_20241231_005d60N_000d19W_LST.csv

-BEGIN HEADER-
NASA/POWER Source Native Resolution Daily Data
Dates (month/day/year): 01/01/2005 through 12/31/2024 in LST
Location: latitude  5.6   longitude -0.19
elevation from MERRA-2: Average for 0.5 x 0.625 degree lat/lon region = 27.76 meters
The value for missing source data that cannot be computed or is outside of the sources availability range: -999
parameter(s):
ALLSKY_SFC_SW_DWN     CERES SYN1deg All Sky Surface Shortwave Downward Irradiance (kW-hr/m^2/day)
T2M                   MERRA-2 Temperature at 2 Meters (C)
RH2M                  MERRA-2 Relative Humidity at 2 Meters (%)
CLOUD_AMT             CERES SYN1deg Cloud Amount (%)
PRECTOTCORR           MERRA-2 Precipitation Corrected (mm/day)
WS2M                  MERRA-2 Wind Speed at 2 Meters (m/s)
-END HEADER-
YEAR,MO,DY,ALLSKY_SFC_SW_DWN,T2M,RH2M,CLOUD_AMT,PRECTOTCORR,WS2M
2005,1,1,5.2625,26.82,85.64,47.32,5.05,1.76
2005,1,2,5.8656,26.32,71.6,12.33,1.08,2.12
2005,1,3,5.7005,26.75,77.16,1.61,0.01,1.35
2005,1,4,5.8726,27.05,77.45,5.2,0.26,1.37
2005,1,5,4.7381,27.02,83.28,51.57,1.25,1.47
2005,1,6,5.0014,27.0,76.57,41.14,0.58,1.66
2005,1,7,4.6589,25.46,69.32,22.14,0.0,2.21
2005,1,8,3.8582,25.03,60.78,27.2,0.0,3.97
2005,1,9,4.8574,24.46,58.97,7.53,0.0,3.91
2005,1,10,5.1818,25.15,59.15,25.34,0.0,3.24
```

Table 2

Study Sites Spanning Ghana's Three Climatic Zones

Site	Climatic zone	Latitude	Longitude
Accra	Coastal savannah	5.60 N	0.19 W
Takoradi	Coastal	4.90 N	1.76 W
Kumasi	Moist semi-deciduous forest	6.69 N	1.62 W
Tamale	Guinea savannah	9.40 N	0.84 W
Navrongo	Sudan savannah	10.89 N	1.09 W

All download scripts, raw and processed data, notebooks, and results are publicly available in a GitHub repository (Adjei-Mensah, 2026; https://github.com/kudumens/capstone-solar-ghana), which already implements the download, validation, and cleaning pipeline. The repository is organized as follows:

```
capstone-solar-ghana/
|-- README.md             (reproduction instructions)
|-- data/raw/             (NASA POWER CSVs, one per site)
|-- data/processed/       (cleaned, feature-engineered datasets)
|-- notebooks/            (01_download ... 07_offgrid_sizing)
|-- src/                  (reusable Python modules)
|-- results/              (metrics tables, figures, sizing tables)
`-- reports/              (interim and final report assets)
```

Data Dictionary

Table 3

Data Dictionary

Variable (POWER code)	Meaning	Type / Units	Role
ALLSKY_SFC_SW_DWN	All-sky global horizontal irradiance	Continuous, $kWh/m^2/day$	Dependent (Y)
T2M	Air temperature at 2 m	Continuous, °C	Predictor
RH2M	Relative humidity at 2 m	Continuous, %	Predictor
CLOUD_AMT	Cloud amount	Continuous, %	Predictor

Variable (POWER code)	Meaning	Type / Units	Role
PRECTOTCORR	Corrected total precipitation	Continuous, mm/day	Predictor
WS2M	Wind speed at 2 m	Continuous, m/s	Predictor
Day-of-year harmonics	Calendar position, cyclically encoded as sine/cosine pairs	Cyclical numeric	Engineered predictor
Season	Wet / dry / Harmattan indicator	Categorical	Predictor / stratifier
Lagged GHI (t − 1 … t − 7)	Recent irradiance history	Continuous, kWh/m²/day	Engineered predictor
Rolling means (7-, 30-day)	Smoothed irradiance trend	Continuous, kWh/m²/day	Engineered predictor
Site / zone	Location identifier	Categorical	Grouping variable
Modeled battery state of charge	Daily energy balance in the RQ4 historical backtest (derived from observed data)	Continuous, kWh	RQ4 state variable

Analytic Approach

Each research question is paired with a testable hypothesis and mapped to statistical and ML techniques. The full analysis is a single pipeline, shown in Figure 2: API ingestion, cleaning and validation, feature engineering (lags, rolling means, cyclical calendar encoding, season indicators), a strict temporal split, baseline models, ML models, statistical comparison, SHAP interpretation, and the off-grid sizing backtest. Tools are Python 3 (pandas, statsmodels, scikit-learn, XGBoost [Chen & Guestrin, 2016], TensorFlow/Keras, and SHAP [Lundberg & Lee, 2017]) with Git and GitHub for reproducibility; all are free and open source. As Figure 3 shows, the record is split chronologically into training (2005–2020), validation (2021–2022), and test (2023–2024) periods with no shuffling; preprocessing parameters (scaling, imputation) are estimated on training data only, and hyperparameters are selected using expanding-window (rolling-origin) time-series cross-validation (Hyndman & Athanasopoulos, 2021). No synthetic or artificially generated data are used anywhere in the study: Every analysis, including the RQ4 sizing, operates exclusively on the observed NASA POWER record, and the only assumed quantities are fixed engineering constants (performance ratio, battery depth of discharge) taken from the published literature.

Figure 2

End-to-End Capstone Pipeline

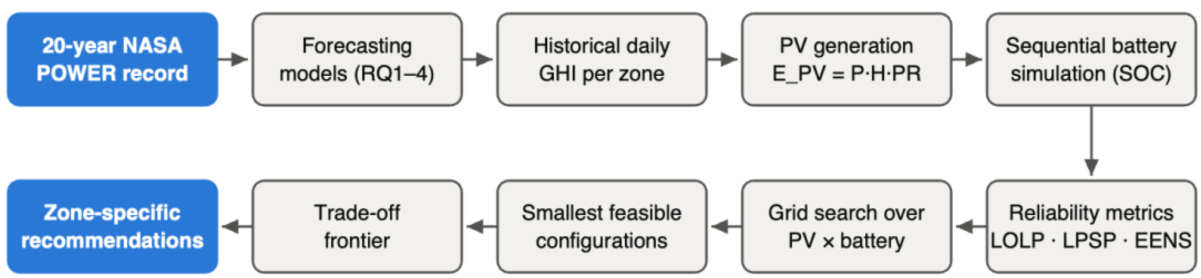
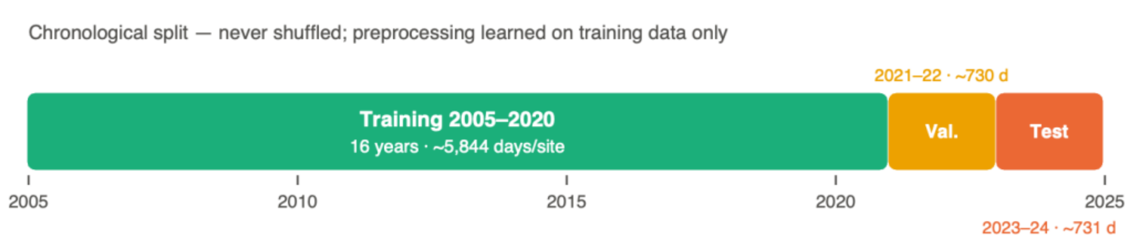


Figure 3

Chronological Train/Validation/Test Split



To prevent look-ahead bias, all predictors used for an h-day-ahead forecast are restricted to values available at or before the forecast origin t: lagged GHI, lagged meteorological variables, and deterministic calendar features. Contemporaneous meteorological values for the target day are not used as forecast inputs because they would not be available at the time of operational forecast issuance; same-day variables are included only in the separate explanatory analysis for RQ2. Formally, the information set is

$$GHI^{\wedge}{}_{t+h} = f(GHI_t, GHI_{t-1}, \dots, X_t, X_{t-1}, \dots, C_{t+h}),\ \ h \in \{1,7\}$$

where X denotes lagged meteorological predictors, and C denotes deterministic calendar features (day-of-year harmonics and season indicators), which are known in advance for any future date.

Research Hypotheses

Each research question is validated by an explicit hypothesis test at $\alpha = .05$. For RQ1, H_0: Skill ≤ 0 (no improvement over the classical benchmark) against H_a: Skill > 0, where Skill = 1 − RMSE_model / RMSE_benchmark, assessed with moving-block-bootstrap confidence intervals on the untouched 2023–2024 test window (Künsch, 1989); RQ1 is otherwise primarily evaluative, with root-mean-square error (RMSE), mean absolute error (MAE), normalized RMSE, and R^2 reported descriptively. For RQ2, for each meteorological driver j, H_0: $\rho_j = 0$ against H_a: $\rho_j \neq 0$, tested with Pearson and Spearman correlations under the Holm correction across the five drivers; relative predictor strength is then ranked with permutation importance and SHAP values, interpreted as model-specific predictive contributions rather than causal effects (Lundberg & Lee, 2017).

For the model-versus-baseline comparison within RQ1, H_0: $E[d_t] = 0$, where d_t is the daily difference in squared forecast errors between the ML model and the classical baseline, against H_a: $E[d_t] < 0$ (the ML model is more accurate), tested with the one-sided Diebold–Mariano test (Diebold & Mariano, 1995) using the Harvey–Leybourne–Newbold (HLN) small-sample correction (Harvey et al., 1997), with a moving-block-bootstrap interval as a dependence-robust check and Holm adjustment across the comparison family. The Diebold–Mariano framework assumes covariance stationarity of the loss-differential series; the HLN correction addresses its small-sample size distortion, and the block length of the moving-block bootstrap will be set on the order of the error series' decorrelation time, with these diagnostics reported alongside the test results. For RQ3, mean absolute errors are compared between wet and dry seasons with a two-sample t test, distributions with the Mann–Whitney U test, and zone-level differences with one-way analysis of variance (ANOVA) or the Kruskal–Wallis test; season boundaries are defined per zone through an explicit month mapping reflecting Ghana's bimodal southern and unimodal northern rainfall regimes. For RQ4, H_0: B_Accra = B_Takoradi = B_Kumasi = B_Tamale = B_Navrongo, where Bz is the minimum battery capacity required in zone z at a fixed reliability target (LOLP ≤ 1%) and the benchmark load, against H_a: at least one zone differs, assessed with block-bootstrap confidence intervals from the 20-year record.

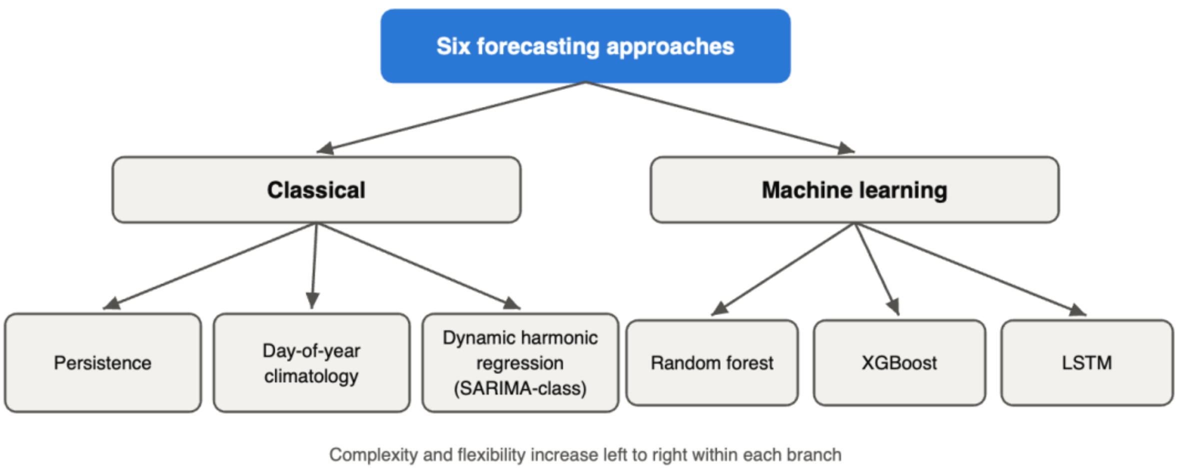
These tests correspond to four directional expectations: H1, ML models significantly outperform the classical benchmarks for 1-day-ahead GHI forecasts in each climatic zone (Skill > 0) and reduce RMSE by at least 15% relative to the day-of-year climatology baseline, with R^2 reported descriptively rather than tested against a fixed threshold; H2, cloud amount and relative humidity show significantly stronger associations with daily GHI than temperature, precipitation, or wind speed; H3, forecast RMSE differs significantly between wet and dry seasons, with the largest degradation in the southern zones during June–August; and H4, required battery capacity for a fixed reliability target differs across climatic zones by at least 25%, implying that uniform national sizing rules materially mis-size systems.

Solution to RQ1–RQ4

RQ1 (statistical and ML). Train random forest (Breiman, 2001), XGBoost (Chen & Guestrin, 2016), and LSTM (Hochreiter & Schmidhuber, 1997) models per site, with a pooled multisite LSTM also evaluated for data efficiency; evaluate on the untouched 2023–2024 test window with RMSE, MAE, normalized RMSE, and R^2, reporting 95% confidence intervals from a moving-block bootstrap over test days (Künsch, 1989). The LSTM uses 30-day input windows, one to two recurrent layers of 32–64 units, dropout, early stopping on the validation window, the Adam optimizer, and five repeated runs with fixed random seeds. Figure 4 shows the full taxonomy of the six forecasting approaches.

Figure 4

Taxonomy of the Six Forecasting Approaches



RQ1, model comparison (statistical). Paired same-day comparison of model errors against the classical benchmarks, i.e., persistence (the forecast equals the last observed value), day-of-year climatology (the forecast equals the historical mean GHI for that calendar day), and a SARIMA-class model implemented as dynamic harmonic regression (Fourier seasonal terms with autoregressive integrated moving-average errors), which handles the approximately 365-day annual cycle more efficiently than a large seasonal SARIMA structure (Hyndman & Athanasopoulos, 2021), using the HLN-corrected Diebold–Mariano test with Holm adjustment and the skill score defined below as the effect-size measure.

RQ2 (statistical and ML). Pearson and Spearman correlations with multicollinearity screening (variance inflation factors), complemented by permutation feature importance and SHAP values on the tree models; hypothesis tests on correlation coefficients at $\alpha = .05$.

RQ3 (statistical). Stratify test-set errors by season (wet, dry, Harmattan) and zone; compare distributions with two-sample t tests, or the Mann–Whitney U test where normality fails; report seasonal skill profiles per zone.

RQ4 (historical backtest and optimization). Drive a daily energy-balance backtest of a PV-plus-battery system through the full 20-year observed irradiance record per zone, as illustrated in Figure 5. Daily PV energy is

$$E_{PV,t} = P_{PV} \times H_t \times PR$$

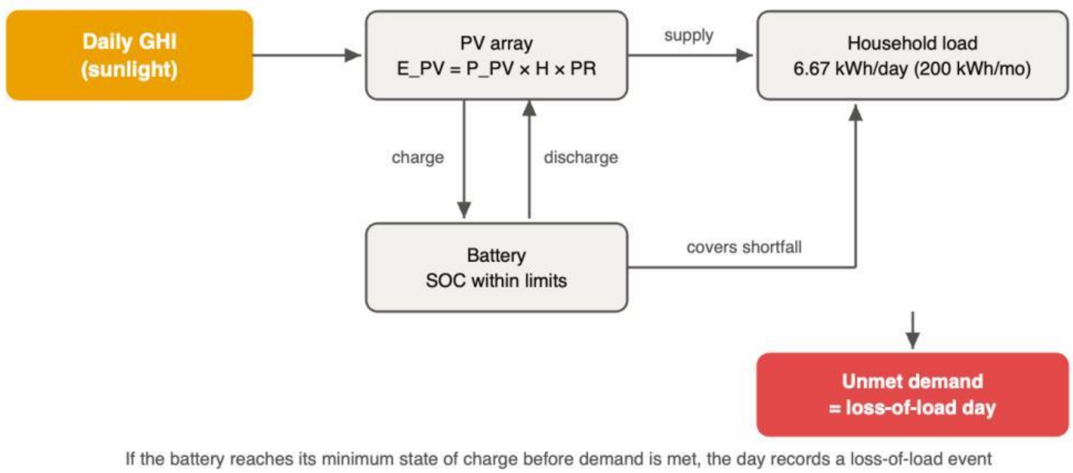
where H_t is the observed daily GHI and PR is the system performance ratio, set to 0.75 with inverter and wiring losses folded in, within the ranges reviewed by Khatib et al. (2013). The battery is modeled with explicit efficiencies and state-of-charge (SOC) limits:

$$SOC_t = min(C_{max}, SOC_{t-1} + \eta_c E_{charge,t} - \frac{E_{discharge,t}}{\eta_d})$$

$$SOC_{min} \leq SOC_t \leq C_{max}, \; SOC_{min} = (1 - DoD_{max})\, C_{max}$$

Figure 5

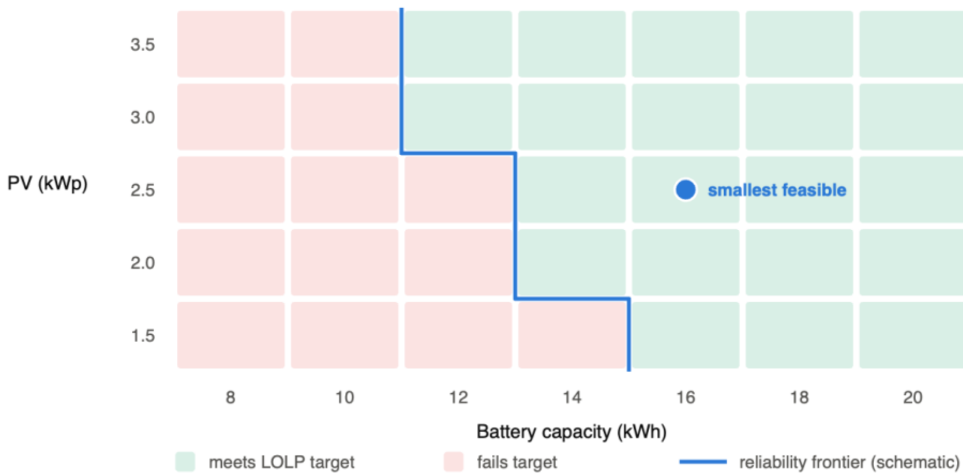
Daily Energy-Balance Flow in the RQ4 Backtest



Charge and discharge efficiencies are $\eta c = \eta d = 0.95$ (approximately 90% round-trip), the maximum depth of discharge (DoD) is set at 0.80 (Khatib et al., 2013), surplus generation is curtailed once the battery is full, and unmet energy is recorded whenever the load exceeds the available supply. The first year of the record (2005) serves as a warm-up period and is excluded from reliability scoring. Battery capacity is treated as constant because the study evaluates initial sizing rather than lifecycle optimization; degradation over the battery's service life (Vetter et al., 2005) is discussed in the Limitations section. The household load is applied as a flat daily profile (200 kWh/month = 6.67 kWh/day), with sensitivity analysis for weekday–weekend and seasonal load variation. The robustness of the sizing results is further assessed by repeating the backtest for performance ratios of 0.70, 0.75, and 0.80. For each candidate configuration (array kWp, battery kWh), reliability is assessed using the three complementary metrics defined below, and a grid search identifies the smallest feasible PV–battery configurations that meet each reliability target (LOLP ≤ 1% and ≤ 5%). Because component prices are outside the study's scope, the results are reported as a PV-versus-battery trade-off frontier, as illustrated in Figure 6, rather than as a single minimum-cost design. Forecast models from RQ1 additionally provide an operational layer: day-ahead low-sun warnings that allow households to shift discretionary loads.

Figure 6

Grid Search Over PV-Array and Battery Capacity With the Reliability Frontier



Evaluation Metrics

Point-forecast accuracy is evaluated with the RMSE, the MAE, the coefficient of determination, and the forecast skill relative to the classical baseline:

$$RMSE = \sqrt{\frac{1}{n} \Sigma_t \, (y_t - \hat{y}_t)^2}$$

$$MAE = \frac{1}{n} \Sigma_t \, |y_t - \hat{y}_t|$$

$$R^2 = 1 - \frac{\Sigma_t \, (y_t - \hat{y}_t)^2}{\Sigma_t \, (y_t - \bar{y})^2}$$

$$Skill = 1 - \frac{RMSE_{model}}{RMSE_{baseline}}$$

Skill is computed against the seasonal-naive baseline, defined here as the day-of-year climatology; skill relative to persistence and dynamic harmonic regression is reported descriptively. System reliability in RQ4 is assessed with three complementary metrics: the LOLP (frequency of failure days), the loss-of-power-supply probability (LPSP; share of total demand that is unserved), and the expected energy not served (EENS; magnitude of the shortfall). N_unmet is the number of observed historical days on which the load could not have been served; a system with one small deficit and one with a full-day outage are thereby distinguished by the LPSP and EENS even when their LOLP is identical:

$$LOLP = \frac{N_{unmet}}{N_{total}}$$

$$LPSP = \frac{\Sigma_t E_{unmet,t}}{\Sigma_t E_{load,t}}$$

$$EENS = \Sigma_t E_{unmet,t}$$

Worked calculation (success criterion). Success on the RQ1 model comparison requires Skill ≥ 0.15. For illustration, if the day-of-year climatology baseline achieves RMSE = 0.80 kWh/m²/day on the test window and XGBoost achieves RMSE = 0.64 kWh/m²/day, then Skill = 1 − 0.64/0.80 = 0.20 ≥ 0.15, satisfying the criterion; the Diebold–Mariano test then determines whether the improvement is statistically significant.

Worked calculation (RQ4 benchmark). A 200 kWh/month household draws 6.67 kWh/day. Sizing on the worst-month resource rather than the annual mean, with worst-month peak sun hours of 4.0 (southern zones, June–August) and PR = 0.75, the required array is 6.67 / (4.0 × 0.75) ≈ 2.2 kWp, rounded upward to six 400-W modules (2.4 kWp); a 2-day-autonomy battery at 80% usable depth of discharge is 2 × 6.67 / 0.8 ≈ 16.7 kWh. These deterministic figures form the starting grid for the LOLP search, which replaces the assumed 2-day autonomy with the empirically observed worst runs of consecutive low-sun days in each zone's 20-year record.

Limitations

Several limitations bound the interpretation of the findings. First, NASA POWER irradiance is satellite-derived; ground-based pyranometer measurements remain the gold standard for solar resource assessment, and although Quansah et al. (2022) support the use of NASA POWER in Ghana, site-level bias and uncertainty are explicitly examined during exploratory analysis rather than assumed away. Second, the sizing backtest runs at daily resolution, so it captures multiday resource variability but not the intraday timing of generation and demand; the resulting capacities are planning estimates rather than installation-ready engineering designs. Third, the household load model is simplified: a flat daily profile with weekday–weekend and seasonal sensitivity scenarios cannot represent every consumption pattern, and sizing results are therefore reported across the 100, 200, and 300 kWh/month tiers. Fourth, the five study sites represent Ghana's climatic zones but do not resolve within-zone microclimates. Finally, battery capacity is held constant over the simulation because the study evaluates initial sizing rather than lifecycle optimization; accounting for degradation over the system lifetime (Vetter et al., 2005) would raise required capacities and is left to future work.

Recommendation and Application

The study targets four user groups. First, off-grid and grid-defecting households, and the solar installers who serve them, receive the headline deliverable: a zone-by-zone sizing table of panels and battery kilowatt-hours required in Accra, Kumasi, Takoradi, Tamale, and Navrongo for the 100, 200, and 300 kWh/month consumption tiers at 99% and 95% reliability, replacing one-size-fits-all autonomy rules with evidence. A household in Navrongo should not pay for the same battery bank as one in Takoradi, and the study will quantify exactly how different their needs are. Second, grid operators (GRIDCo, ECG) receive day-ahead and week-ahead irradiance forecasts with quantified zone- and season-specific skill levels, informing dispatch and reserve planning as solar penetration approaches the Renewable Energy Master Plan targets. Third, developers and financiers receive zone-level profiles of resource reliability and forecastability for siting and bankability assessment. Fourth, policymakers (the Energy Commission of Ghana) receive evidence on where solar output is most predictable and where off-grid economics are strongest, informing siting incentives and rural electrification strategy. All outputs are released as a reproducible open pipeline that can be rerun for any West African location covered by NASA POWER.

Adoption of these outputs is expected to differ by user group, and practical implementation challenges are acknowledged. The sizing tables are designed to be read directly from a one-page, zone-by-zone lookup, so installers need no code or statistical training to use them; dissemination will therefore target installer networks and the Energy Commission of Ghana's renewable-energy programs, supported by brief nontechnical documentation included in the repository. Institutional users such as GRIDCo and ECG can rerun the open pipeline at no licensing cost, but operational adoption would require validating the forecasts against their own telemetry, and NASA POWER's multiday data latency limits real-time use; the forecasts are accordingly positioned for planning and reserve scheduling rather than real-time dispatch.

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